

hey create a prompt example for chain of thoughts (COT)-for creating a Automation Engineer for infrastructure engineering

Absolutely. For a **Chain-of-Thought (CoT)-style prompting example**, it's better to ask the AI to provide a **structured reasoning summary and actionable steps** rather than requesting its private hidden chain of thought.

Example Prompt — Automation Engineer for Infrastructure Engineering

CoT-Style Prompt: Infrastructure Automation Engineer

Role:

You are an expert Infrastructure Automation Architect and Senior Automation Engineer with 15+ years of experience in enterprise IT infrastructure, backup, storage, monitoring, cloud, Python automation, REST APIs, Playwright, PyAutoGUI, and DevOps.

Objective:

I want to develop myself into a highly skilled **Infrastructure Automation Engineer** who can automate repetitive infrastructure operations and build reliable automation solutions for enterprise environments.

Task:

Create a structured learning and career-development roadmap for becoming an Infrastructure Automation Engineer.

Think through the problem systematically and provide a **concise reasoning summary** for each major decision.

Step 1 — Assess the Role

Explain:

- What an Infrastructure Automation Engineer does
- Daily responsibilities
- Important technical skills
- Difference between Infrastructure Engineer, Automation Engineer, and DevOps Engineer
- Skills required for an L2/L3-level engineer

Step 2 — Identify the Technology Stack

Build a technology roadmap covering:

Programming

- Python
- Bash/PowerShell
- Exception handling
- Logging
- File handling
- JSON/YAML
- Object-oriented programming

Automation

- REST API automation
- Python Requests
- Playwright
- PyAutoGUI
- Selenium
- SSH automation
- Task scheduling

Infrastructure

- Servers
- Windows/Linux
- Networking fundamentals
- Storage
- Backup technologies
- Monitoring
- Virtualization
- Cloud

DevOps

- Git/GitHub
- CI/CD
- Docker
- Ansible
- Terraform
- Jenkins/GitHub Actions

AI + Automation

- Generative AI
- LLM APIs
- Prompt engineering
- AI-assisted troubleshooting
- AI agents
- RAG
- Local LLMs such as Ollama

Step 3 — Create a 90-Day Learning Plan

Create a **12-week roadmap**.

For every week provide:

1. Topic to learn
2. Why it is important
3. Tools to install
4. Hands-on exercises
5. Infrastructure automation use case
6. Mini project
7. Expected outcome
8. Interview questions

Step 4 — Build Real-World Projects

Design at least **10 enterprise-style automation projects**, such as:

- Backup health-check automation
- Storage capacity monitoring
- Server health monitoring
- Automated backup failure analysis
- REST API-based infrastructure reporting
- Automated ticket creation
- Daily infrastructure status report
- Log analysis automation
- Automated compliance report
- AI-powered infrastructure troubleshooting assistant

For each project explain:

- Business problem
- Architecture
- Python components

- APIs required
- Automation workflow
- Error handling
- Logging
- Security considerations
- Expected result

Step 5 — Design a Capstone Project

Create one advanced project called:

"AI-Powered Infrastructure Automation & Monitoring Platform"

The platform should:

User → Web UI/API → Python Automation Engine → REST APIs/SSH → Infrastructure Systems → Monitoring/Logs → AI Analysis → Report/Alert

Define:

- Architecture
- Components
- Python project structure
- API endpoints
- Database requirements
- Authentication
- Logging
- Error handling
- Scheduling
- AI integration
- Dashboard
- Deployment approach

Step 6 — Create a Python Project Structure

Provide a professional enterprise folder structure such as:

```
infrastructure-automation/  
|  
├─ app/  
|   ├─ api/  
|   ├─ automation/  
|   ├─ monitoring/  
|   └─ backup/
```

```
|   ├── storage/
|   ├── ai/
|   └── utils/
|
|── tests/
|── config/
|── logs/
|── reports/
|── scripts/
|── requirements.txt
|── .env
|── .gitignore
└── README.md
```

Explain the purpose of each directory.

Step 7 — Career Preparation

Create:

- Infrastructure Automation Engineer skill matrix
- L2/L3 interview preparation
- Python interview questions
- REST API interview questions
- Automation framework questions
- Infrastructure troubleshooting questions
- DevOps questions
- AI automation questions
- Real-world scenario-based interview questions

Step 8 — Final Skill Assessment

At the end, create a scoring matrix:

Skill	Beginner	Intermediate	Advanced	My Target
Python				
REST API				
Playwright				
PyAutoGUI				

Linux

Backup

Storage

Git

Docker

Ansible

Cloud

GenAI

Finally, provide a **90-day action plan**, showing exactly what I should practice each day.

Do not provide hidden/private chain-of-thought. Instead, provide concise reasoning summaries, assumptions, decision criteria, and actionable steps for each stage.

Simple CoT Prompt Formula

You can also remember this structure for your assignments:

Role → Goal → Context → Break into Steps → Reasoning Summary → Execute → Validate → Improve → Final Output

For example:

"Act as a Senior Infrastructure Automation Architect. Help me become an Infrastructure Automation Engineer. Break the goal into logical stages, explain the purpose and decision criteria for each stage, identify the required technologies, create hands-on projects, validate the learning path against real enterprise requirements, and finish with a 90-day roadmap and skill assessment. Do not reveal private chain-of-thought; provide concise reasoning summaries instead."

This is a strong **Sequential + CoT-style prompt** for your GenAI learning/assignment work.

ChatGPT can make mistakes. Check important info.